
Bracket Norm Is Not Mechanism: Validity Conditions for Noncommutative Perturbation Scores

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Abstract

Direction instability—the mean pairwise angular deviation of perturbation signatures across contexts—predicts drug mechanism transport across cell lines. We ask what this prediction *licenses*. Through pre-registered experiments across three perturbation domains—8,949 LINCS L1000 compounds (978 transcriptomic features), 1,676 Perturb-seq gene knockdowns (Replogle et al., 2022) (2,073 genes, of which 1,567 have DepMap essentiality labels), and 25,254 JUMP-CP compounds (3,180 Cell Painting morphological features)—we show that validity conditions do not reclassify perturbations: they determine what claims a practitioner can make about the ranking. In all three domains, correction preserves the ranking (LINCS: $\rho > 0.99$; Perturb-seq: $\rho = 0.91$; JUMP-CP: $\rho = 0.99$). Three drug-domain validity conditions are tested empirically: phenotype-projected bracket correlates with on-target genetic perturbation connectivity ($\rho = 0.38$, $p < 10^{-27}$) while raw bracket does not ($\rho = -0.04$); transport-stable bracket outpredicts raw bracket for held-out cell-line effects in all 5 cross-validation folds; and localization to target-gene knockdown signatures improves prediction for intracellular enzyme inhibitors but not receptor-mediated drugs, producing an informative null. Permutation baselines confirm that the transport-stable advantage is drug-specific rather than an algebraic artifact of Fréchet variance coupling ($p_{\text{perm}} < 10^{-3}$, variance-preserving signature shuffle) and that essential-gene divergence in Perturb-seq exceeds background K562/RPE1 divergence ($p_{\text{perm}} < 10^{-4}$, label-permutation null). Item-level rank shifts identify specific confound-dominated perturbations—proteasome subunits and spliceosome components shift 660–900 positions after essential-gene correction—providing actionable flags for individual scores. Pre-registered scorecard: 3 of 5 drug hypotheses confirmed (H2, H3, H5), 1 refuted with artifact pass (H1), 1 informative null (H4). The validity ladder functions as a proof system for mechanistic claims: each rung does not change the score, it upgrades the inferential warrant attached to that score.

1 Introduction

The Connectivity Map paradigm (Lamb et al., 2006; Subramanian et al., 2017) established that transcriptomic-signature similarity predicts drug mechanism of action. Direction instability extends this from pairwise similarity to angular consistency: rather than asking whether two drugs produce similar signatures, it asks whether a single drug’s signature rotates across cell lines. Tower (2026) applied this metric to 8,949 drugs in LINCS L1000, showing that mechanism-conserving compounds (pan-HDAC inhibitors, topoisomerase inhibitors, CDK inhibitors) achieve low direction instability while receptor-mediated drugs cluster near the population mean of 0.92. The HDAC selectivity gradient—pan-isoform inhibitors at $D = 0.56$ – 0.63 , selective inhibitors at $D = 0.96$ – 0.99 —provides a natural controlled comparison within a single drug class.

These results establish that direction instability captures genuine biological mechanism conservation. They leave open a different question: under what conditions does a direction-instability score license a mechanistic *claim*? A score of $D = 0.55$ means signatures point in similar directions across cell lines. It does not, by

itself, mean the drug acts through a specific pathway, that the effect is therapeutic rather than cytotoxic, or that the observation will replicate in a new dataset.

Confound corrections for perturbation scores are well-established: CERES corrects copy-number artifacts in CRISPR screens (Meyers et al., 2017); connectivity-metric evaluations benchmark signature-similarity scores (Li et al., 2020). These corrections improve the *accuracy* of the score. We ask a different question: when does a score—even an accurate one—license a mechanistic *claim*? A correction that preserves the global ranking ($\rho > 0.99$) adds no discriminative power, yet it upgrades the inferential warrant by ruling out a specific confounding explanation.

We formalize this through a validity ladder: a sequence of conditions, each building on the previous, that progressively strengthen what a direction-instability measurement licenses. The ladder has seven rungs, from declaring the process view (what counts as a context?) through phenotype alignment and cross-context transport to holdout prediction. Each rung answers a specific threat to interpretation. The ladder is a proof system, analogous to the Bradford Hill criteria for causal inference and to recent construct-validity frameworks for ML benchmarks (Freiesleben & Zezulka, 2025): satisfying more rungs does not change the number, it changes what you are allowed to conclude from it.

This paper tests whether validity conditions matter empirically across three perturbation domains. In LINCS L1000, pre-registered experiments probe specific failure modes of drug direction instability: cytotoxicity masquerading as mechanism conservation and broad therapeutics that are smooth rather than mechanism-absent. In Perturb-seq genetic knockdowns from Replogle et al. (Replogle et al., 2022), we test whether essential-gene correction—the genetic analogue of toxicity correction—changes the Grassmannian geodesic distance ranking between K562 and RPE1 perturbation response subspaces. In JUMP-CP Cell Painting morphological profiles (Chandrasekaran et al., 2023), we test whether removing cell-health features—the morphological analogue of toxicity and essentiality correction—changes the direction-instability ranking across 25,254 compounds measured in ≥ 5 imaging sources. We report honest nulls alongside confirmations, because the pattern of where validity conditions bite—and where they do not—is itself the finding.

2 Background

2.1 Direction instability

For a drug tested in K cell lines, let $\mathbf{s}_1, \dots, \mathbf{s}_K \in \mathbb{R}^{978}$ denote its consensus signature vectors. Define unit-normalized signatures $\hat{\mathbf{s}}_i = \mathbf{s}_i / \|\mathbf{s}_i\|_2$. Direction instability is:

$$D = 1 - \frac{2}{K(K-1)} \sum_{i < j} \hat{\mathbf{s}}_i^\top \hat{\mathbf{s}}_j \quad (1)$$

$D = 0$ when all signatures are collinear. $D = 1$ when orthogonal on average. (The term “bracket norm” in the companion paper (Tower, 2026) refers to this same quantity viewed as a noncommutative perturbation score; we use “direction instability” throughout to avoid terminological collision with unrelated norm constructions.) This metric was introduced in Tower (2026) to characterize drug mechanism transport and was shown to anticorrelate with gene-level Jaccard overlap of top perturbed genes (Spearman $\rho = -0.79$).

2.2 Grassmannian geodesic distance in Perturb-seq

Direction instability generalizes from vectors to subspaces. For a gene knocked down in two cell types, let $U_1, U_2 \in \text{Gr}(k, d)$ denote the top- k PCA subspaces of the perturbation response in each cell type. The Grassmannian geodesic distance $\delta(U_1, U_2) = \|\boldsymbol{\theta}\|_2$, where $\boldsymbol{\theta}$ are the principal angles between the subspaces, measures how much the perturbation response *rotates* across cell types. With $K = 2$ contexts, this reduces to a single pairwise comparison—the subspace analogue of direction instability for a drug tested in two cell lines.

2.3 Direction instability in Cell Painting morphological profiles

Cell Painting (Chandrasekaran et al., 2023) measures 3,180 morphological features per well (cell area, shape, granularity, texture, and intensity across five fluorescent channels: DNA, RNA, ER, AGP, Mito). JUMP-CP profiles 116,000 compounds across 10 imaging sources. Direction instability applies directly: each compound’s morphological signature in a given source is a vector in $\mathbb{R}^{3180}$, and D measures angular consistency across sources. A subset of features—cell area, compactness, eccentricity, DNA/Mito intensity, granularity, and radial distribution—reflect general cell health rather than compound-specific morphological effects. The cell-health correction removes these 743 features before recomputing direction instability, analogous to the toxicity correction in LINCS and the essential-gene correction in Perturb-seq.

2.4 The validity problem

Consider two drugs with identical direction instability, $D = 0.55$:

- **Drug A** is a pan-HDAC inhibitor. Its conserved signature reflects chromatin remodeling genes (*SUV39H1*, *MYC*, *CDK6*). The consistency comes from targeting a universally expressed enzymatic family.
- **Drug B** is a mitochondrial toxin. Its conserved signature reflects a generic stress response (*HSPA1A*, *DDIT3*, *ATF4*). The consistency comes from triggering the same damage pathway regardless of cell identity.

Both achieve $D = 0.55$. The interpretation differs entirely: Drug A transports a therapeutic mechanism; Drug B transports cellular violence. The raw metric cannot distinguish these cases. A validity condition—
toxicity correction, in this instance—can.

2.5 The validity ladder

We define seven rungs:

1. **Process view declared.** What constitutes a “context”? (Cell lines, cell types, patients, time points.)
2. **Reliable vector field.** Signatures have adequate signal-to-noise for direction to be meaningful.
3. **Perturbation estimable.** The effect can be isolated from background variation.
4. **Localized to biology.** Instability concentrates in pathway-relevant genes, not globally.
5. **Phenotype-aligned.** The direction of conservation matches a known target pathway.
6. **Transports across contexts.** The score replicates in held-out contexts (low Fréchet variance).
7. **Predicts holdout.** The metric predicts an external outcome in new data.

Rungs 1–3 are prerequisites for any directional metric; they are satisfied by design in the datasets we use (declared cell-line contexts, replicated signatures, perturbation effects above background) and are not tested further. Rungs 4–6 are the validity conditions we test empirically: each eliminates a specific interpretive threat without altering the underlying score. Rung 7 (holdout prediction in a genuinely new dataset) is not tested here: Experiment 5’s leave-one-cell-line-out cross-validation tests Rung 6 (transport across contexts within LINCS), not generalization to an independent dataset.

3 Related work

CERES (Meyers et al., 2017) corrects copy-number artifacts in CRISPR screens; integrated Broad-Sanger analyses (Pacini et al., 2021) document cross-study divergence. These corrections operate on dependency-score magnitudes; our contribution is orthogonal, correcting subspace geometry and showing that rank-preserving corrections carry inferential value. The Connectivity Map framework (Subramanian et al., 2017; Lamb et al., 2006) and subsequent metric evaluations (Li et al., 2020) establish the transcriptomic-similarity paradigm that direction instability extends. LEMUR (Ahlmann-Eltze & Huber, 2025) performs multivariate regression on Grassmann manifolds for multi-condition Perturb-seq; our Experiment 7 uses a simpler Grassmannian geodesic distance. The validity ladder draws on construct-validity frameworks for computational scores (Freiesleben & Zezulka, 2025; Watson et al., 2025) and the SBB principles for perturbation-prediction models (SBB Consortium, 2026).

4 Methods

4.1 Data

Perturbation signatures were drawn from LINCS L1000 Phase I (GEO accession GSE92742; ~ 1.3 million profiles across ~ 50 cell lines and $\sim 20,000$ compounds), using Level 5 replicate-collapsed z-scores across 978 landmark genes (Subramanian et al., 2017). We retained 8,949 small-molecule perturbagens tested in ≥ 5 cell lines, the same dataset characterized in Tower (2026).

Perturb-seq data derive from Replogle et al. (Replogle et al., 2022): genome-scale CRISPRi knockdowns in K562 (chronic myelogenous leukemia) and RPE1 (retinal pigment epithelium) cells. We extracted 1,676 genes with ≥ 30 perturbed cells in both cell types, computed per-gene expression responses over 2,073 highly variable genes, and fitted $k = 5$ PCA subspaces per gene per cell type. (Of these 1,676, 1,567 have matching DepMap Chronos labels and are used for the HP3 essentiality analysis; all 1,676 are used for HP1 and HP2.) Grassmannian geodesic distances were computed between the K562 and RPE1 subspaces for each gene. Both the K562 and RPE1 screens use guide libraries targeting DepMap-essential genes (Replogle et al., 2022); 85% of RPE1 targets overlap with K562-essential targets (1,875 of 2,204). The “non-essential” comparison group therefore consists of genes that were selected for essentiality but fall below the stringent Chronos < -0.5 threshold in one or both cell types. Essentiality labels derive from DepMap Chronos scores (Dempster et al., 2021): of 1,567 genes with both geodesic distances and DepMap labels, 855 with Chronos < -0.5 in both K562 and RPE1 were classified as universal-essential. A label-permutation null (shuffling essential/non-essential labels while holding geodesic distances fixed, 10,000 permutations) tested whether the observed group difference exceeds background K562/RPE1 divergence.

JUMP-CP Cell Painting profiles derive from the Cell Painting Gallery (Chandrasekaran et al., 2023): variance-stabilized, MAD-normalized compound profiles across 10 imaging sources. The full dataset contains 116,750 compound profiles; after deduplication and filtering to compounds present in ≥ 5 sources, 25,254 compounds remain (78% attrition, primarily from compounds tested in < 5 sources). We aggregated per compound $\times$ source by mean and computed direction instability over the full 3,180 feature set. Cell-health features (743 features matching Area, Compactness, Eccentricity, DNA/Mito Intensity, Granularity, Radial Distribution, and Texture Entropy) were identified by substring matching on CellProfiler feature names. Corrected direction instability was computed on the remaining 2,437 non-health features.

All scoring functions and curated drug lists were committed before any real LINCS signatures were analyzed. Perturb-seq hypotheses (HP1–HP3) were pre-registered before the correction analysis was run; the integrity boundary, decision criteria, and commit SHA are documented in the code repository.¹

Test family and multiplicity. We treat the five LINCS drug hypotheses (H1–H5) as one test family and the three Perturb-seq hypotheses (HP1–HP3) as a second. Within each family, hypotheses test distinct

¹Code and pre-registration: <https://github.com/elliotttower/direction-instability-drug-validity>. All experiments, scoring functions, and permutation-null scripts are included. LINCS L1000 data derive from GEO accession GSE92742; Perturb-seq data from Replogle et al. (2022); JUMP-CP data from Chandrasekaran et al. (2023); DepMap Chronos scores from Dempster et al. (2021).

validity conditions rather than the same quantity at multiple thresholds, so we pre-registered individual decision criteria ($\alpha = 0.0125$ for HP1–HP3; specific thresholds for H1–H5) rather than a family-wise correction. The pre-registered success criterion—3 of 5 drug hypotheses confirmed—accounts for multiplicity at the study level. After applying Holm–Bonferroni correction within the drug family, H3 ($p < 10^{-27}$) and H5 (5/5 folds, $p_{\text{perm}} < 10^{-3}$) survive at $\alpha = 0.05$; H2 ($p = 0.28$, binomial 7/10) and H4 (gap = 0.018) do not. H1’s criterion-pass is an artifact (see Results). Within Perturb-seq, HP3 ($p < 10^{-33}$, $p_{\text{perm}} < 10^{-4}$) survives; HP1 and HP2 are characterization tests (ranking correlations) without formal hypothesis tests.

4.2 Phenotype-projected bracket (Experiment 3)

Rung 5 of the validity ladder requires that direction instability align with a known target pathway. We test this using genetic perturbation connectivity: for each drug with an annotated target gene (Drug Repurposing Hub, 1,551 drugs with target annotations), the “on-target direction” is the consensus shRNA knockdown signature of that target gene in LINCS L1000. The shRNA signatures span 20 cell lines, dominated by a core set (PC3, VCAP, MCF7, A375, HA1E, A549, HT29, HCC515, HEPG2; 9 lines contributing >95% of signatures). Each consensus averages across all cell lines and hairpins for genes with ≥ 3 shRNA constructs (4,369 target genes eligible).

Phenotype-projected bracket projects the instability vector onto this on-target direction. On-target enrichment is quantified by cosine squared between each drug’s mean signature and its target gene’s shRNA consensus direction—the fraction of mean-signature variance aligned with the genetic perturbation.

Pre-registered prediction (H3): Spearman $|\rho| > 0.3$ between phenotype-projected bracket and on-target enrichment, AND $|\rho| < 0.15$ for raw bracket versus the same enrichment.

4.3 Transport-stable bracket (Experiment 5)

Rung 6 requires that the score replicate across held-out contexts. Transport-stable bracket penalizes raw direction instability by Fréchet variance across contexts. We evaluate via leave-one-cell-line-out cross-validation over the first 5 cell lines alphabetically (A375, A549, A673, AGS, ASC). For each fold, we compute raw and transport-stable bracket on the training cell lines and measure cosine similarity between the held-out cell line’s signature and the training consensus direction. This held-out outcome matches the direction-based estimand of direction instability: a drug whose signatures are directionally consistent in training should produce a held-out signature aligned with the training consensus.

Pre-registered prediction (H5): Spearman correlation between transport-stable bracket and held-out cosine > correlation for raw bracket, in at least 4 of 5 folds.

4.4 Essential-gene correction on Perturb-seq (Experiment 7)

Universal-essential knockdowns (DepMap Chronos < -0.5 in both K562 and RPE1) may inflate or deflate Grassmannian geodesic distances through a shared lethal-response subspace. We computed the Fréchet mean subspace of the 855 essential-gene perturbation responses (separately in each cell type) via iterative projection on $\text{Gr}(5, 2073)$. We then projected this essential-response subspace out of each gene’s response subspace, re-orthogonalized, and recomputed geodesic distances on the residual.

Pre-registered prediction (HP1): The essential-response correction changes the geodesic-distance ranking. Decision criterion: Spearman ρ between raw and corrected distances below 0.70 = “bites”; 0.70 – 0.83 = “marginal”; ≥ 0.83 = “doesn’t bite” (the HDAC selectivity gradient gives $\rho = 0.83$ after drug toxicity correction).

Pre-registered characterization (HP3): Universal-essential knockdowns produce higher geodesic distances than non-essential knockdowns (Cohen’s $d > 0$), consistent with the failure-mode inversion identified before freezing.

Table 1: Leave-one-cell-line-out cross-validation. Transport-stable (TS) bracket outpredicts raw bracket in all 5 folds. Raw bracket is anti-predictive: higher raw instability in training predicts *lower* held-out cosine consistency, while higher TS bracket predicts higher consistency.

Held-out cell	N_{drugs}	ρ_{raw}	ρ_{TS}	Winner
A375	8,328	-0.480	-0.155	TS
A549	8,888	-0.494	-0.201	TS
A673	357	-0.725	+0.725	TS
AGS	356	-0.712	+0.714	TS
ASC	2,345	-0.365	+0.346	TS

4.5 Supplementary experiments

Three additional drug-domain experiments and one morphological experiment are reported in full in the Appendix. **Experiment 1** (toxicity correction) tests whether cytotoxic drugs masquerade as mechanism-conserving via shared stress programs (H1). **Experiment 2** (broad-mechanism drugs) tests whether broad-target therapeutics fall below the population median direction instability (H2). **Experiment 4** (localization score) tests whether restricting direction instability to target-gene-relevant genes improves MOA prediction (H4). **Experiment 8** (JUMP-CP cell-health correction) tests whether removing cell-health features from Cell Painting profiles changes the direction-instability ranking. We also report HP2 (pathway function prediction from corrected Perturb-seq distances) in the Appendix.

5 Results

5.1 Experiment 3: Phenotype-projected bracket separates on-target from off-target instability

Among 795 drugs with annotated targets and matched shRNA consensus signatures, phenotype-projected bracket correlates with on-target enrichment at Spearman $\rho = 0.376$ ($p = 4.8 \times 10^{-28}$). Raw bracket shows no correlation with the same enrichment ($\rho = -0.043$, $p = 0.22$). Both pre-registered criteria are satisfied: $|\rho_{\text{projected}}| = 0.376 > 0.3$ and $|\rho_{\text{raw}}| = 0.043 < 0.15$.

Interpretation. H3 confirms Rung 5 of the validity ladder empirically. Raw direction instability measures whether a drug’s signatures point in consistent directions across cell lines, but this consistency could arise from any coherent program—on-target, off-target, or stress-related. Projecting instability onto the genetic perturbation direction of the drug’s annotated target isolates the mechanistically relevant component. The near-zero correlation for raw bracket shows that global instability carries no information about on-target alignment; the strong correlation for projected bracket shows that the on-target projection recovers it.

The gene-set construction avoids researcher degrees of freedom: the shRNA consensus signature is an empirical object from the same LINCS L1000 platform, requiring no pathway database selection or gene-set size decisions. H3 and H4 share this shRNA dependency (documented in DEVIATION_LOG.md entry 3) and are not independent tests.

5.2 Experiment 5: Transport-stable bracket outpredicts raw bracket in held-out cell lines

Transport-stable bracket outpredicts raw bracket in all 5 cross-validation folds (Table 1). Raw bracket is *anti-predictive*: its Spearman correlation with held-out cosine consistency is negative in every fold, reaching $\rho = -0.73$ in the A673 fold. Transport-stable bracket is positively predictive in all folds, reaching $\rho = +0.72$ in the same fold. The pre-registered criterion (transport-stable wins in ≥ 4 of 5 folds) is satisfied at 5/5.

The effect is strongest in rare cell lines. Folds holding out rare cell lines (A673: 357 drugs, AGS: 356 drugs) show the largest absolute correlations and the starkest sign flips. In these folds, removing

a single cell line from training substantially changes the consensus direction, and drugs with high raw instability (inconsistent directions in training) produce held-out signatures that point away from the training consensus. Transport-stable bracket corrects for this by penalizing high Fréchet variance, so drugs whose training instability is driven by a single outlier cell line are downweighted.

Common cell lines show the same direction, smaller magnitude. In the A375 and A549 folds (8,000+ drugs), raw bracket remains anti-predictive ($\rho \approx -0.49$) and transport-stable bracket improves over raw, though both remain negative. The consensus direction is stable when computed from many cell lines, so the held-out prediction task is easier and both metrics perform more similarly—but the ordering is preserved.

Shared-variance caveat. The transport-stable penalty and the held-out cosine outcome both depend on the training-set signature geometry, raising a potential circularity concern: both quantities share variance through the training consensus direction. The sign reversal in rare-cell-line folds (raw anti-predictive, TS positively predictive) argues against a trivial shared-variance explanation—if the improvement were purely algebraic, we would expect monotonic improvement rather than a sign reversal. A variance-preserving permutation null tests this directly: held-out *signatures* are shuffled across drugs while each drug’s training data (raw DI, Fréchet variance, TS, and consensus vector) remains fixed, and held-out cosines are recomputed against each drug’s original consensus. This preserves the mechanical coupling between consensus tightness and cosine magnitude. Under this null (1,000 permutations), TS wins only 0.1 of 5 folds on average with mean $\Delta\rho = -0.12$ (raw DI actually outpredicts TS when drug-specific signal is broken). The observed advantage (5/5 folds, $\Delta\rho = +0.84$) falls entirely outside the null distribution ($p_{\text{perm}} < 10^{-3}$), confirming that the Fréchet penalty rewards genuine drug-specific structure rather than exploiting algebraic coupling with the held-out outcome.

5.3 Experiment 7: Essential-gene correction preserves ranking but reveals confound-dominated machinery

HP3: Failure-mode inversion. Universal-essential knockdowns have *higher* Grassmannian geodesic distances than non-essential knockdowns (mean 2.80 vs. 2.67, Cohen’s $d = +0.60$, Mann–Whitney $p < 10^{-33}$; $n = 1,567$ genes with both geodesic distances and DepMap labels, of which 855 universal-essential and 712 non-essential). Essential genes do not produce a shared response across cell types; they *diverge*—the same gene’s lethal disruption manifests through different transcriptomic programs in K562 and RPE1. This inverts the predicted failure mode: in drugs, “violence” (cytotoxicity) was predicted to mimic transport but does not; in genetics, “violence” (essential knockdowns) produces the opposite of transport.

A permutation null (shuffling essential/non-essential labels across the 1,567 genes while holding geodesic distances fixed, 10,000 permutations) confirms that this divergence exceeds background K562/RPE1 divergence: the observed difference (0.123) falls outside the null 95% interval $[-0.021, 0.022]$, $p_{\text{perm}} < 10^{-4}$. The null embeds the baseline cross-cell-type divergence into both groups by preserving the geodesic distance distribution; what it breaks is only the essential/non-essential label assignment. That the observed effect exceeds the null by $\sim 6\times$ the null half-width rules out the confound that both screens’ DepMap-essential gene libraries (85% overlap) mechanically drive the separation.

HP1: Correction does not bite. After projecting out the essential-response Fréchet mean subspace, Spearman ρ between raw and corrected geodesic distances is 0.91 ($p < 10^{-300}$). By the pre-registered criterion, this “doesn’t bite”—the same qualitative outcome as the drug HDAC gradient ($\rho = 0.83$). The global ranking is preserved.

Item-level rank shifts are the primary finding. The essential-gene correction’s value is not at the population level (where $\rho = 0.91$ confirms stability) but at the item level: individual genes shift hundreds of rank positions, identifying which transport scores are most confound-dominated. The top 100 rank-shifters are enriched for spliceosomal complex assembly (Fisher’s exact OR = 8.9, $p = 1.3 \times 10^{-7}$) and proteasomal catabolic processes (OR = 4.1, $p = 6.7 \times 10^{-5}$): proteasome subunits (*PSMA3*, *PSMB6*, *PSMC4*) shift 750–900 positions; spliceosome components (*SF3B1*, *SNRPG*, *DDX23*) shift 660–820 positions. These core-machinery complexes are the genes whose cross-cell-type geometry is most dominated by the shared essential-

response axis. The correction provides a concrete action: flag these genes before interpreting their transport scores as mechanistically meaningful.

5.4 Summary of supplementary experiments

Full methods, tables, and discussion for Experiments 1, 2, 4, 8, and HP2 appear in the Appendix.

H1 (toxicity correction): Refuted with artifact pass. Among 13 cytotoxic drugs present in the dataset, most are already high-instability ($D > 0.87$ for 9 of 13). Toxicity correction shifts values by < 0.01 . The predicted failure mode—cytotoxic drugs masquerading as mechanism-conserving via shared stress programs—does not hold: cytotoxicity produces genuinely context-dependent signatures in 978-gene space.

H2 (broad-mechanism drugs): Confirmed with nuance. Seven of 10 broad-mechanism drugs fall below the population median ($D = 0.92$), confirming H2. Pan-HDAC inhibitors dominate the low end ($D = 0.49$ – 0.55). The pre-registered second prediction—that context-specific kinase inhibitors would fall above the median—failed. The sorting axis is target *breadth* (number of gene programs engaged), not clinical specificity.

H4 (localization): Informative null. Macro-averaged AUROC gap between localization score and raw bracket is $+0.018$ (below the $+0.05$ threshold). Localization improves prediction for intracellular enzyme inhibitors (topoisomerase: $+0.49$, CDK: $+0.26$) but hurts for receptor-mediated drugs (acetylcholine: -0.20 , PPAR: -0.19). Rung 4 is domain-dependent.

Experiment 8 (JUMP-CP cell-health correction): Doesn’t bite. Among 25,254 compounds, corrected direction instability tracks raw at $\rho = 0.987$. Despite 743 health features constituting 23% of the feature space—proportionally six times larger than the drug toxicity gene set (41/978, 4%)—the ranking impact is smaller, strengthening the thesis that health features carry less cross-source variance than mechanism-specific features.

HP2 (pathway function prediction): Doesn’t bite. GO Biological Process Jaccard overlap has near-zero correlation with geodesic distance ($|\rho| < 0.025$) for both raw and corrected forms. Geometric similarity of perturbation responses is distinct from functional pathway relatedness.

6 Discussion

6.1 Validity conditions license claims, they do not reclassify

The central finding is negative in the most useful sense: validity conditions do not reorganize the direction-instability ranking. This thesis was genuinely at risk: had any correction produced $\rho < 0.70$ while simultaneously improving prediction of an external outcome, the “reclassify” interpretation would have prevailed—the raw ranking would have been wrong, not merely under-warranted. That no correction crossed this threshold across three domains is the empirical result, not a definitional guarantee. In LINCS drugs, the HDAC selectivity gradient is preserved after toxicity correction ($\rho = 0.83$; the global 8,949-drug ranking gives $\rho > 0.99$). In Perturb-seq, the essential-gene correction preserves the global ranking ($\rho = 0.91$). In JUMP-CP, the cell-health correction leaves the ranking nearly unchanged ($\rho = 0.99$). The Perturb-seq result involves a degenerate $K = 2$ pairwise comparison rather than the genuine $K \geq 5$ angular-consistency computation used in LINCS and JUMP-CP; the qualitative conclusion holds across domains, but the Perturb-seq domain tests a weaker geometric estimand.

What changes is the inferential warrant. Phenotype projection (H3) proves that direction instability decomposes into on-target and off-target components ($\rho = 0.38$ vs. -0.04). Transport stability (H5) proves that the metric predicts held-out cell-line effects only after Fréchet variance correction; raw bracket is anti-predictive. Essential-gene correction identifies proteasome/spliceosome genes as confound-dominated while

Table 2: Cross-domain summary. The correction-vs-raw Spearman ρ quantifies how much the domain-specific validity condition reorganizes the ranking. All three domains exceed the pre-registered “doesn’t bite” threshold ($\rho \geq 0.83$). The LINCS HDAC gradient ρ measures preservation of the fine-grained selectivity ordering within a single drug class (20 compounds); the global 8,949-drug $\rho > 0.99$.

Domain	Correction	Features	N	ρ
LINCS L1000 (HDAC gradient)	Toxicity genes	978	20	0.83
LINCS L1000 (global)	Toxicity genes	978	8,949	>0.99
Perturb-seq	Essential-gene subspace	2,073	1,676	0.91
JUMP-CP	Cell-health features	3,180	25,254	0.99

the population ranking holds. The validity ladder is a proof system: each rung eliminates a specific threat to interpretation. A practitioner who reports $D = 0.55$ after toxicity correction, pathway localization, and held-out replication makes a stronger claim than one who reports raw $D = 0.55$. The number is the same; the license differs. This aligns with recent work on construct validity for computational scores (Freiesleben & Zezulka, 2025; Watson et al., 2025).

6.2 Cross-domain consistency

The same geometric object—angular deviation of perturbation-response directions across contexts—captures mechanism conservation in three unrelated domains (Table 2). All three exhibit the same pattern: the relevant validity condition preserves the global ranking while identifying individual perturbations whose scores are most confound-dominated.

The domain-specific failure modes differ in mechanism: in drugs, cytotoxicity produces high-instability signatures (context-dependent stress cascades); in genetics, universal-essential knockdowns produce high-distance subspaces (divergent lethal programs); in morphology, cell-health features contribute a small fraction of the cross-source variance. All three invert the naive prediction that “violence mimics transport.”

6.3 Scope and limitations

Direction instability measures angular consistency of perturbation signatures across contexts. It does not measure therapeutic efficacy, in-vivo translatability, or clinical specificity. A drug with low direction instability and full ladder validation has a strong claim to *mechanism-conserving transcriptomic effects across cell lines in vitro*; whether that mechanism is therapeutically relevant is a question the ladder does not answer.

The drug experiments H3–H5 depend on shRNA knockdown signatures as ground-truth on-target directions, assuming correct target annotations. Drugs with unannotated or incorrectly annotated targets (7,398 of 8,949) are excluded. LINCS L1000 measures 978 landmark genes, restricting the toxicity gene set to the landmark panel. The Perturb-seq analysis uses $K = 2$ cell types, collapsing to a single pairwise comparison with no Fréchet variance or transport stability test. Both screens target DepMap-essential genes; a genome-wide screen with genuine non-essential controls would provide a sharper HP3 test, though the label-permutation null ($p_{\text{perm}} < 10^{-4}$) rules out the library-overlap confound directly. JUMP-CP cell-health features were defined by substring matching rather than an independent biological criterion.

7 Conclusion

Direction instability is a reliable measure of perturbation-response consistency across contexts, whether applied to drug signatures in 978-gene transcriptomic space, genetic knockdown responses as subspaces on the Grassmannian, or compound morphological profiles in 3,180-feature Cell Painting space. Validity conditions do not improve the metric’s ability to rank perturbations—the ranking is already biologically meaningful in all three domains tested ($\rho > 0.99$ for LINCS, 0.91 for Perturb-seq, and 0.99 for JUMP-CP).

after correction). They improve the practitioner’s ability to make claims about that ranking. Phenotype projection separates on-target from off-target instability ($\rho = 0.38$ vs. -0.04); transport-stable bracket improves over raw bracket in all 5 cross-validation folds ($p_{\text{perm}} < 10^{-3}$), reversing raw’s anti-prediction in rare cell lines and attenuating it in common ones; localization helps for intracellular enzyme targets but not receptor-mediated drugs, constraining where Rung 4 applies. Each condition eliminates a specific interpretive threat: toxicity correction rules out stress-mediated false positives; essential-gene correction rules out shared lethal-response geometry; cell-health correction rules out viability-driven consistency. The domain-specific failure modes differ, but the pattern is the same: correction preserves population rankings while identifying individual perturbations most susceptible to confounding. The validity ladder is a proof system for mechanistic inference from geometric perturbation scores.

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A Supplementary experiments

A.1 Experiment 1: Toxicity correction has negligible effect in LINCS

Among the 16 pre-registered cytotoxic compounds, 13 were present in the ≥ 5 -cell-line dataset. Their raw direction instabilities ranged from 0.47 to 0.93 (Table 3). This immediately contradicts the predicted failure mode: cytotoxic drugs are not false-positive transporters. They are already high-instability—their signatures rotate across cell lines rather than conserving a consistent stress direction.

After removing stress/apoptosis gene axes, direction instability shifted by < 0.01 for all 13 drugs. The correction has no practical effect because the premise of the failure mode—that cytotoxic drugs achieve low bracket via a shared stress program—does not hold in LINCS L1000 data.

Table 3: Direction instability for known cytotoxic drugs. Most are above the population median (0.92), contradicting the predicted failure mode. Correction for stress genes shifts values by < 0.01 .

Drug	D_{raw}	D_{corr}	Δ
staurosporine	0.47	0.48	+0.01
doxorubicin	0.74	0.74	0.00
etoposide	0.78	0.78	0.00
camptothecin	0.82	0.82	0.00
vincristine	0.87	0.87	0.00
paclitaxel	0.89	0.89	0.00
cisplatin	0.91	0.91	0.00
tunicamycin	0.92	0.92	0.00
thapsigargin	0.93	0.93	0.00
<i>Population median</i>	0.92	—	—

H1 passes its pre-registered criterion (85% of toxic drugs fall below the corrected median), but this is an artifact of the criterion design: most toxic drugs are already high-instability, so correction moves them negligibly rather than reclassifying them. We classify H1 as *refuted with artifact pass*: the predicted mechanism (cytotoxic drugs masquerading as mechanism-conserving via shared stress programs) is wrong. Cytotoxicity produces genuinely context-dependent signatures.

Methods. We curated 16 known cytotoxic compounds (staurosporine, camptothecin, doxorubicin, etoposide, cisplatin, vincristine, paclitaxel, actinomycin-d, thapsigargin, tunicamycin, brefeldin-a, rotenone, antimycin-a, oligomycin-a, menadione, hydrogen-peroxide) and 41 stress/apoptosis marker genes (*HSPA1A*, *BAX*, *TP53*, *CASP3*, *ATF4*, *DDIT3*, etc.). Toxicity-corrected bracket removes these gene axes from the signature matrix before recomputing direction instability on the residual. **Pre-registered prediction (H1):** $\geq 60\%$ of cytotoxic drugs move from the top 20% of raw bracket to below-median toxicity-corrected bracket.

A.2 Experiment 2: Broad-mechanism drugs confirm; specificity gradient is weaker than predicted

Seven of 10 broad-mechanism drugs fell below the population median direction instability of 0.92, confirming H2 (Table 4). The strongest signal came from pan-HDAC inhibitors: vorinostat ($D = 0.49$) and trichostatin-a ($D = 0.55$) are among the most mechanism-conserving drugs in the entire dataset.

Table 4: Direction instability for pre-registered broad-mechanism drugs. 7/10 fall below the population median (0.92). Pan-HDAC inhibitors dominate the low end.

Drug	Rationale	D	Below median?
vorinostat	pan-HDAC	0.49	Yes
trichostatin-a	pan-HDAC	0.55	Yes
sirolimus	mTOR	0.78	Yes
methotrexate	DHFR	0.81	Yes
dexamethasone	GR agonist	0.83	Yes
metformin	AMPK	0.86	Yes
lovastatin	HMGCR	0.88	Yes
atorvastatin	HMGCR	0.91	No
simvastatin	HMGCR	0.93	No
aspirin	COX	0.96	No

The pre-registered second prediction—that context-specific kinase inhibitors would fall above the median—failed. All 8 tested kinase inhibitors had direction instability below the population median (0.73–0.91; Table 5). The sorting axis is target *breadth*—how many gene programs a compound engages—not target *specificity*.

Table 5: Direction instability for pre-registered context-specific drugs. Contrary to prediction, all fall below the population median.

Drug	Target	D	Above median?
vemurafenib	BRAF V600E	0.73	No
imatinib	BCR-ABL	0.78	No
erlotinib	EGFR	0.81	No
crizotinib	ALK/ROS1	0.83	No
lapatinib	HER2	0.84	No
gefitinib	EGFR	0.86	No
PCI-34051	HDAC8	0.89	No
tubastatin-a	HDAC6	0.91	No

Methods. We curated 10 broad-mechanism drugs: statins (atorvastatin, simvastatin, lovastatin), metformin, dexamethasone, sirolimus, methotrexate, aspirin, and pan-HDAC inhibitors (vorinostat, trichostatin-a). We also curated 8 context-specific drugs (vemurafenib, imatinib, erlotinib, crizotinib, lapatinib, gefitinib, PCI-34051, tubastatin-a). **Pre-registered prediction (H2):** $\geq 60\%$ of broad-mechanism drugs fall below the population median direction instability.

A.3 Experiment 4: Localization predicts intracellular but not receptor-mediated mechanism

The macro-averaged one-vs-rest AUROC gap between localization score and raw bracket is +0.018 (95% bootstrap CI $[-0.010, +0.048]$), below the pre-registered threshold of +0.05. Per-MOA AUROC gaps reveal a consistent pattern: localization improves prediction for drugs targeting intracellular enzymes and hurts for receptor-mediated drugs (Table 6).

Table 6: Per-MOA AUROC gaps (localization – raw) for the 15 eligible MOA classes. Intracellular enzyme targets show positive gaps; receptor-mediated targets show negative gaps.

MOA class	N	AUROC gap
topoisomerase inhibitor	15	+0.491
CDK inhibitor	16	+0.258
mTOR inhibitor	13	+0.176
p38 MAPK inhibitor	10	+0.165
HDAC inhibitor	20	+0.076
glucocorticoid receptor agonist	27	+0.061
MEK inhibitor	13	+0.036
adrenergic receptor agonist	14	−0.036
calcium channel blocker	11	−0.087
cyclooxygenase inhibitor	17	−0.099
EGFR inhibitor	39	−0.107
histamine receptor antagonist	26	−0.135
estrogen receptor agonist	16	−0.148
PPAR receptor agonist	16	−0.188
acetylcholine receptor antagonist	23	−0.201

Methods. For each drug, the region mask comprises the 100 genes with largest absolute z -score in the target gene’s shRNA consensus signature ($\sim 10\%$ of the 978 landmark genes). The localization score is the ratio of pathway-restricted bracket to global bracket. MOA prediction uses macro-averaged one-vs-rest AUROC across MOA classes with ≥ 10 drugs (15 classes, 223 eligible drugs). **Pre-registered prediction (H4):** Macro-averaged AUROC for MOA prediction from localization score $>$ AUROC from raw bracket by at least 0.05.

A.4 Experiment 8: Cell-health correction has negligible effect in JUMP-CP

Among 25,254 compounds tested in ≥ 5 imaging sources, raw direction instability has population mean 0.92 and median 0.95 (left-skewed by a tail of compounds with low cross-source consistency). After removing 743 cell-health features, corrected direction instability tracks the raw ranking at Spearman $\rho = 0.987$ ($p < 10^{-300}$). By the pre-registered criterion, this “doesn’t bite.” The 743 health features constitute 23% of the 3,180-feature space—proportionally six times larger than the drug toxicity gene set (41/978, 4%)—yet produce less ranking impact.

A.5 Pathway function prediction (HP2)

GO Biological Process Jaccard overlap has near-zero correlation with geodesic distance in both raw ($\rho = 0.024$, $p < 10^{-7}$) and corrected ($\rho = -0.002$, $p = 0.68$) forms. The correction removes the tiny positive relationship that existed rather than strengthening it (improvement = -0.022). Geodesic distance measures transcriptomic-geometry rotation, which is orthogonal to pathway co-membership at this scale.